

ShopFlow AWS Project

Lambda Design

Serverless Functions - SQS Triggers - Email - Order Processing

Module 14 of 18

Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
Author	Manish Kumar Sharma
Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

AWS Lambda runs event-driven functions without managing any servers. ShopFlow uses Lambda for asynchronous tasks that don't require always-on compute: sending transactional emails and processing order events. Lambda functions are triggered by SQS queues and scale automatically from zero to thousands of parallel executions.

2. Functions

Function Name	Runtime	Trigger	Timeout	Memory	Purpose
shopflow-email-notifier-dev	Java 21	shopflow-notifications SQS (batch 10)	60s	512 MB	Send transactional emails via SendGrid
shopflow-order-processor-dev	Java 21	shopflow-order-events SQS (batch 5)	60s	512 MB	Update inventory, trigger fulfillment

3. SQS Trigger Configuration

Setting	Email Notifier	Order Processor
Batch Size	10 messages per invocation	5 messages per invocation
Batching Window	5 seconds	10 seconds
Partial Batch Failure	ReportBatchItemFailures	ReportBatchItemFailures

ReportBatchItemFailures: if a batch of 10 has 2 failures, only those 2 messages are re-queued. The other 8 are deleted. Without this, all 10 would be retried, causing duplicate sends.

4. VPC Configuration

Component	Value
Security Group	shopflow-lambda-sg
Subnets	shopflow-prv-1a + shopflow-prv-1b
Outbound	Port 3306 to RDS, port 443 to external APIs (HTTPS)

5. Cold Start

Cold Start Phase	Duration	What Happens
Container provision	~100ms	AWS allocates a container for the function
JVM startup	~500ms	JVM initializes, class loading
Handler init	~1-2s	Spring context loads (if using Spring Cloud Function)
First invocation	~2-3s total	Subsequent calls reuse warm container (~50ms)

6. Reserved Concurrency

Function	Reserved Concurrency	Reason
Email notifier	10	Prevent SQS burst from consuming all account concurrency
Order processor	5	DB connection pool limits -- max 5 parallel DB writers